Task1: NoSQL Fundamentals

Task A (Document Design): Design a JSON document for a Product. Include: Name, Category, price, stock, tags.

Aim:
To store product information in a structured JSON format in MongoDB, enabling efficient querying, filtering, and management of product data.

```
db.products.insertMany([
 {
   name: "Wireless Mouse",
category: "Electronics",
   price: 799,
stock: 120,
   tags: ["wireless", "accessories", "computer"]
 },
 {
   name: "Bluetooth Headphones",
   category: "Electronics",
price: 2499,    stock: 60,
   tags: ["audio", "wireless", "headphones"]
 },
 {
   name: "OLice Chair",
category: "Furniture",
price: 5999,    stock:
25,
   tags: ["chair", "oLice", "comfort"]
 },
 {
   name: "Notebook Pack",
category: "Stationery",
   price: 299,
stock: 200,
   tags: ["paper", "writing", "school"]
 },
```

```
 {
   name: "Smart Water Bottle",
category: "Lifestyle",
   price: 1999,    stock: 40,    tags:
["fitness", "smart", "hydration"]
}
})
```

Explanation:
This query inserts multiple product documents into the products collection in MongoDB. Each document contains name, category, price, stock, and tags fields, providing structured information about products. The tags array allows multiple keywords per product for easier searching and categorization.

```
_id: ObjectId('6971ff394bf1b2054a9da7c9')
name : "Wireless Mouse"
category : "Electronics"
price : 799
stock : 120
▸ tags : Array (3)
```

```
_id: ObjectId('6971ff394bf1b2054a9da7ca')
name : "Bluetooth Headphones"
category : "Electronics"
price : 2499
stock : 60
▸ tags : Array (3)
```

```
_id: ObjectId('6971ff394bf1b2054a9da7cb')
name : "Office Chair"
category : "Furniture"
price : 5999
stock : 25
▸ tags : Array (3)
```

```
_id: ObjectId('6971ff394bf1b2054a9da7cc')
name : "Notebook Pack"
category : "Stationery"
price : 299
stock : 200
▸ tags : Array (3)
```

```
_id: ObjectId('6971ff394bf1b2054a9da7cd')
name : "Smart Water Bottle"
category : "Lifestyle"
price : 1999
stock : 40
▸ tags : Array (3)
```

Task B (Embedding vs Referencing): Should "Reviews" be embedded in the Audiobook document or kept separate? Justify your choice for a read-heavy app.

Reviews should be added to the main documents only if they are summarised (small) else all of them must be stored in diFerent documents and reference must be given to their respective main documents.